The Economics of Hosting Mega Sporting Events

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Abstract

Cities and countries that bid for the Olympics or the World Cup usually promise that the event will pay for itself through tourism, jobs, and new infrastructure. To test that promise, I built a dataset of 91 international sporting events held between 1988 and 2025 and compared each event's cost with its revenue, the number of venues it built, and the host country's economic growth before and after the event. Only 22 of the 91 events earned more revenue than they cost. Construction was the dividing line: 20 of the 32 events that built no new venues earned more than they cost, and none of the 32 events that built five or more did. Published studies and audits show the same problem at the budget stage: according to the Oxford Olympics Study, the Olympics have gone over budget by an average of 156 percent in real terms. Host countries' growth usually slowed after their events, but the countries that bid for the same events and lost slowed by about as much, so the data do not show that hosting itself helped or hurt growth. Cases from Rio de Janeiro, Seoul, Tokyo, and PyeongChang show that a venue stays useful only when it has regular users after the event.

Author's Note

I am a student in the Academy of Finance at Fort Lee High School, and I wanted to understand how money and major sporting events are connected: whether the events that cities and countries compete to host actually pay for themselves. I started this project in 2025. I collected the figures in Google Sheets and used Python and RStudio for the calculations and the charts. Brian Song, a biostatistician at Johns Hopkins Hospital, discussed the project with me and reviewed an earlier draft. His main advice was that more data and more numbers would make the paper stronger and more analytical, and I followed it by building the 91-event dataset and by testing the patterns I found in it. He is a reviewer, not a coauthor. The dataset and the calculations are posted with this paper at jades259.github.io/jadeseo.

1. Introduction

Hosting a major international sports event such as the Olympic Games or the FIFA World Cup is usually presented as a chance for a city or country to grow its economy, attract tourists, create jobs, and improve its infrastructure. However, these events also require governments to invest heavily in facilities, transportation, and security systems, and because many of these costs and benefits continue for years after the closing ceremony, the economic consequences of hosting are difficult to measure. This paper looks at the financial results of past international sporting events and asks whether hosting actually pays off for the countries that do it.

Five questions guide this paper, and Sections 3 and 4 answer them in the same order. The first is whether events earn back what they cost. The second is whether hosts spend what they said they would: before an event, governments and organizing committees publish budgets based on projected construction costs, operating expenses, tourism, and revenue, and I compared those budgets with final costs, using published studies and government audits, to see whether hosts kept their promises. The third is how new venues affect an event's finances and what happens to those venues afterward, since a new venue can generate activity for a few weeks but may carry maintenance and operating costs for decades, while other venues continue to serve their communities through sports, concerts, and other events. The fourth is whether hosting helps the wider economy through faster GDP growth or new jobs. The fifth is opportunity cost: money spent on an event is money that could have gone to schools, hospitals, public transportation, or affordable housing. This paper does not estimate what any government would actually have done with that money, because that alternative is never observed; instead, it measures how large a shortfall hosts had to cover and examines one case, Rio de Janeiro, in which cuts to public services happened at the same time as Olympic spending.

To answer these questions, I compared events by estimated costs, revenues, reported profits and losses, numbers of new venues, reported jobs, and the GDP growth of host countries and the countries that bid and lost, before and after each event. Because the dataset covers many host countries over several decades, it can show patterns that a study of a single event would miss, and those patterns are what a government should weigh before deciding whether to bid.

2. Data and Methods

2.1 The dataset

I built a dataset of 91 international sporting events from existing public sources. The events took place between 1988 and 2025. They include 18 editions of the Africa Cup of Nations (AFCON), 15 of Copa América (one of them the 2016 Copa América Centenario), 10 of the AFC Asian Cup, and nine each of the Summer Olympics, the Winter Olympics, the FIFA World Cup, the UEFA European Championship (the Euros), and the Asian Games, along with three single-venue events held in 2025: the UEFA Champions League Final in Munich, The Open golf championship at Royal Portrush in Northern Ireland, and the Wimbledon tennis championships. The Appendix lists all 91 events. For each event, I recorded the sport, the host country, the year, the estimated cost, the revenue, the reported profit or loss, the number of new or renovated facilities, the estimated number of jobs created, the host country's unemployment rate, and two GDP values: the host country's average annual GDP growth rate over the five years before the event and over the five years after it. I first recorded single-year growth rates from Worldometer's country GDP pages (Worldometer, n.d.), which report International Monetary Fund data, and then rebuilt both five-year averages from the annual real GDP growth rates in the IMF's World Economic Outlook database (International Monetary Fund, 2026), so that one unusual year could not decide the result. For Rio de Janeiro 2016, for example, the before average covers 2011 to 2015 and the after average covers 2017 to 2021. For events with more than one host country, I used the average of the hosts' rates, and I took unemployment rates from the World Bank's World Development Indicators (World Bank, n.d.); they are recorded in the dataset but not analyzed in this paper. One event, The Open in 2025, reported its profit only as a range, so it is left out of the profit counts. Where the information was available, I also recorded the cities or countries that bid for the event and lost.

2.2 Measures

Because events report profit using different accounting methods, I relied mainly on the revenue-to-cost ratio, which divides an event's revenue by its estimated cost; a ratio above 1.0 means that the event earned more than it cost. To examine the relationship between construction and financial results, I sorted events into three groups by the number of new venues they built (none, one to four, and five or more) and compared the median ratio and the share of profitable

events in each group. To measure employment, I divided each event's reported jobs by its estimated cost in millions of dollars, which gives the number of jobs created per $1 million spent. All amounts are in U.S. dollars of each event's own year and are not adjusted for inflation; costs reported in euros or pounds were converted to dollars.

I chose real GDP growth because it summarizes the performance of a whole economy. For each host country, I subtracted the five-year average before the event from the five-year average after it. The main comparison uses the 77 events held from 1988 to 2019, because for the 11 events held from 2020 to 2024 the five years after the event are not complete, and for the three 2025 events they have not begun. A host's growth can change for reasons that have nothing to do with the event, so I also compared each host with the countries that bid for the same event and lost, over the same years: for each event, I calculated the average change for its losing bidders, counting each country in a joint bid separately, and subtracted it from the host's change. I counted a country as a losing bidder only if it bid for that edition and was not chosen. I therefore left out countries that had been chosen to host and then withdrew or were replaced. Several of them lost the event because of a war or a crisis that would also distort their growth; Libya, for example, was chosen to host the Africa Cup of Nations in 2013 and again in 2017 and lost both because of armed conflict. I also left out countries that won a different edition in the same vote. This comparison was possible for 55 of the 77 events. The other 22 had no usable comparison. Some had no losing bidders, because tournaments such as Copa América rotate among their member countries. In others, the only countries listed did not count as losing bidders under this rule, or the only losing bidder has no IMF data, as with the Soviet Union for the 1990 World Cup. A few other bidders without IMF data, such as Yugoslavia, were left out of their events' averages. To judge whether a split between increases and decreases could be due to chance, I used a sign test. A sign test only counts how many events went up and how many went down, and it asks how often a split that uneven would happen if up and down were equally likely, like flipping a coin. I report medians because a few countries had extreme changes in growth, such as Libya after its 2011 civil war. I also report a 95 percent confidence interval for the median difference between hosts and bidders, which I estimated by resampling the 55 events thousands of times (a bootstrap). As Section 5 explains, many forces other than a sporting event move GDP, so this comparison can show whether growth changed but not why.

2.3 Supplementary sources

The dataset records each event's estimated cost, but in most cases it does not record the original projection made when the host bid for the event. To compare projected and actual costs, I therefore used published studies and government audits for Olympic Games from 1976 to 2020, mainly the Oxford Olympics Study (Flyvbjerg et al., 2016), a report by the United Kingdom's National Audit Office on London 2012 (National Audit Office, 2007), and the final cost figures released by the Tokyo 2020 organizing committee (Agence France-Presse, 2022). The sources for the 91-event dataset itself are listed separately at the end of the References. To study what happened to venues after the events, I used the International Olympic Committee's 2022 announcement of its study of post-Games venue use (International Olympic Committee, 2022) and news reports on specific venues in Brazil and South Korea, along with figures published by the Tokyo Metropolitan Government. Section 3.8 also draws on South Korean government sources and news reports about the Seoul Olympics and the 2002 World Cup.

3. Results

3.1 Most events do not recover their costs

The first result is that most events did not recover their costs. Of the 90 events with profit data, 28 reported a profit, and only 22 of the 91 events earned revenue greater than their estimated cost. The two numbers differ because a reported profit is usually the organizing committee's operating surplus, which leaves out construction and other public spending, so an event can report a profit even when its revenue is smaller than its total cost. Together, the 91 events had estimated costs of about $589.6 billion and revenue of about $96.2 billion, which means that they earned back roughly 16 cents for every dollar they cost. The 2022 World Cup in Qatar accounts for $220 billion of the $589.6 billion in costs, or about 37 percent, and as Section 3.7 explains, much of that figure is national infrastructure rather than tournament spending. Without Qatar, the other 90 events had costs of about $369.6 billion and revenue of about $88.7 billion, so they earned back about 24 cents for every dollar, which is better but still far from breaking even. For this reason, the rest of the paper relies mainly on medians, which outliers like Qatar cannot pull far. The size of the event also made a large difference, although not in a straight line. As Table 1 shows, events that cost less than $1 billion recovered a median of about 85 percent of their costs through revenue, whether they cost under $100 million or between $100

million and $1 billion. For events that cost $1 billion to $10 billion, the median fell to 48 percent, and above $10 billion it fell to only 13 percent. Once an event costs more than about $1 billion, it rarely pays for itself, and recent Summer Olympics and World Cups all fall into the most expensive groups.

Table 1. Revenue-to-cost ratio by event cost

Event cost	Number of events	Median revenue-to-cost ratio
Under $100 million	25	0.84
$100 million to under $1 billion	25	0.87
$1 billion to $10 billion	27	0.48
Over $10 billion	14	0.13

Note: The two events that cost exactly $1 billion, AFCON 2010 and AFCON 2023, are counted in the $1 billion to $10 billion group.

3.2 Projected and actual costs

Budgets made at the bidding stage almost always underestimate final costs. The Oxford Olympics Study, which compared projected and actual sports-related costs for the Games held from 1960 to 2016, found that every Games with available data went over budget, with an average overrun of 156 percent. The average overrun was 176 percent for the Summer Games and 142 percent for the Winter Games (Flyvbjerg et al., 2016). Table 2 shows several examples, together with the more recent case of Tokyo 2020.

Table 2. Projected and actual costs at selected Olympic Games

Games	How the cost was measured	Increase over projection	Source
Montreal 1976	Sports-related cost, real terms	720%	Flyvbjerg et al. (2016)
Barcelona 1992	Sports-related cost, real terms	266%	Flyvbjerg et al. (2016)
Athens 2004	Sports-related cost, real terms	49%	Flyvbjerg et al. (2016)
Beijing 2008	Sports-related cost, real terms	2%	Flyvbjerg et al. (2016)
London 2012	Sports-related cost, real terms	76%	Flyvbjerg et al. (2016)
Sochi 2014	Sports-related cost, real terms	289%	Flyvbjerg et al. (2016)
Rio 2016	Projected cost at the time of the study	51%	Flyvbjerg et al. (2016)
Tokyo 2020	¥734 billion in the 2013 bid file; ¥1.4238 trillion final	About 94%	Agence France-Presse (2022)

Note: The Oxford study notes that China's official figures may be less reliable than those of other hosts, so the Beijing result should be read with caution. The Tokyo percentage is my calculation from the two published figures; part of that increase came from the one-year postponement caused by COVID-19.

Different sources measure cost in different ways, and that difference is part of the problem. For London 2012, the National Audit Office reported that the bid had estimated a gross cost of just over £4 billion, but the public funding package announced in March 2007 had grown

to more than £9 billion (National Audit Office, 2007). That increase is much larger than the 76 percent that the Oxford study calculated, because the Oxford study counts only sports-related costs while the funding package also covers infrastructure, security, and contingency. Qatar shows the same pattern in its stadium budget. Its World Cup bid proposed about $4 billion for stadiums, but reported stadium costs reached $6.5 billion to $10 billion (Killingstad, 2022). Every Games in the Oxford study went over budget, so a host should expect its bid budget to rise.

3.3 New venues and financial results

Construction was the strongest pattern in the data. Events held in existing stadiums and facilities performed much better than events that had to build venues from the ground up, as Table 3 shows.

Table 3. Financial results by number of new venues built

New venues built	Number of events	Median revenue-to-cost ratio	Revenue greater than cost	Reported a profit
None	32	1.09	20 of 32 (63%)	18 of 31 (58%)
One to four	27	0.66	2 of 27 (7%)	5 of 27 (19%)
Five or more	32	0.32	0 of 32 (0%)	5 of 32 (16%)

Note: The Open (2025) reported its profit only as a range, so it is not counted in the profit column.

An event that built no new venues earned a median of $1.09 for every dollar it cost, and 58 percent of those events reported a profit. Events that built one to four venues earned a median of 66 cents for every dollar, and only 19 percent were profitable, while events that built five or more earned a median of 32 cents, and only 16 percent were profitable. The clearest difference is therefore between building no new venues and building any. The gap is even sharper when revenue is compared directly with cost: 20 of the 32 events that built no new venues earned more than they cost, compared with 2 of the 27 that built one to four and none of the 32 that built five or more. Differences this large are very unlikely to be due to chance (a Kruskal-Wallis test, which compares groups by rank instead of by average, gives $p < 0.001$). By type of event, single-venue events and the Euros had the strongest ratios, while the Summer Olympics and the World Cup had the weakest. This pattern does not prove that construction causes losses, because the events that build the most are also the largest and most complex, and the three groups also differ in the kinds of events they contain, so part of the gap may reflect event type rather than construction. Part of the gap is also built into the measure, because the cost of a new venue is

counted in the ratio while the money the venue earns after the event is not. To check this, I compared only the continental soccer tournaments (AFCON, Copa América, the AFC Asian Cup, and the Euros), which share a similar format: the 28 that built no new venues had a median ratio of 1.08, while the 24 that built at least one had a median of 0.62 (a Mann-Whitney test gives $p < 0.001$), so the gap holds within a single kind of event. The data do show, however, that the hosts that relied on facilities they already had were far more likely to break even.

Figure 1 places every event on one chart. Events that built no new venues cluster around or above the break-even line, while events that built five or more sit well below it, and the shortfall grows as costs rise from left to right.

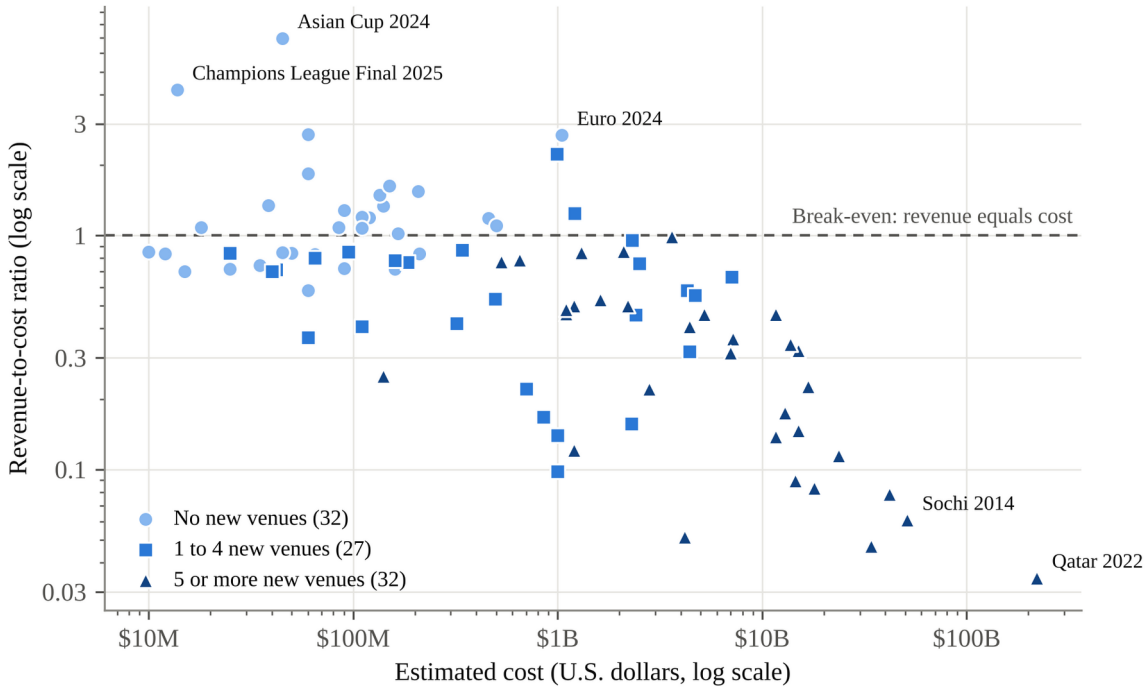


Figure 1. Estimated cost and revenue-to-cost ratio for all 91 events, by number of new venues built. Both axes use a logarithmic scale; the dashed line marks break-even.

3.4 What happens to venues after the event

Counting new venues shows how much a host built, but not whether the buildings were useful afterward, so I also looked at what happened to venues once the events were over. The International Olympic Committee's study of 817 permanent and 106 temporary venues used from 1896 to 2018 reported that 85 percent of permanent Olympic venues are still in use, a share that rises to 92 percent for the 206 permanent venues used in the 21st century (International

Olympic Committee, 2022). That figure comes from the organization that awards the Games, however, and a venue counted as “in use” may host events every day or only a few times a year.

Rio de Janeiro shows what can go wrong. Six months after the 2016 Games, four of the new arenas in the main Olympic Park had failed to find private-sector management and had passed over to the federal government. The Maracanã stadium had lost its electricity in a dispute over about $1 million in unpaid bills, and only 260 of the 3,604 apartments in the $1 billion Athletes Village had been sold (Wade, 2017). Brazil’s 2014 World Cup left a similar legacy in Brasília, where the $550 million Mané Garrincha stadium, the most expensive stadium of the tournament, was being used as a parking lot for buses within a year (NPR, 2015). Germany avoided this problem at Euro 2024 by holding every match in existing stadiums that already had professional clubs as tenants, so there were no new venues left to fill after the tournament.

3.5 GDP growth before and after the event

Host countries usually grew more slowly after their events than before them. Real GDP growth slowed after 49 of the 77 events and rose after 28. The median host grew 3.6 percent a year in the five years before its event and 2.4 percent a year in the five years after it, and the median change, calculated event by event, was −0.8 percentage points. A sign test shows that a split this uneven would be unlikely to happen by chance ($p = 0.02$), and the result holds when each host, or group of co-hosts, is counted only once, even though Brazil, China, and the United States each hosted four of these events: growth slowed for 33 of the 49 hosts ($p = 0.02$). The slowdown looked larger where there was more construction, with median changes of −0.3 points for events with no new venues, −0.8 for events with one to four, and −1.1 for events with five or more, but these differences are small enough to be due to chance (a Kruskal-Wallis test gives $p = 0.52$).

On its own, that pattern makes hosting look harmful, but the comparison with losing bidders points to a different explanation. Over the same years, the countries that bid for the same events and lost also slowed. Across the 55 events with a comparison, hosts slowed by a median of 0.5 percentage points and their losing bidders by 0.3. The median host did about 0.3 points worse than its losing bidders, but hosts did better in 26 of the 55 events and worse in 29, a split that chance explains easily ($p = 0.79$), and the 95 percent confidence interval for the median difference runs from about −0.9 to +0.5 points. Figure 2 shows the events falling on both sides of the line where hosts and bidders change by the same amount. For the Olympics and the World

Cup alone, hosts did worse than their losing bidders in 14 of 23 events, but that gap is also too small to rule out chance (p = 0.40). One possible explanation for the shared slowdown is that countries tend to bid when their economies are strong, and strong periods are usually followed by slower ones. Another is that building for an event raises a host's growth in the years before it, which would make the later slowdown look larger. The data therefore do not show that hosting made a country's economy grow faster or slower than it would have without the event, although with 55 events they cannot rule out a small effect in either direction. This echoes the finding of Rose and Spiegel (2009) that countries that bid for the Olympics and lost gained as much in trade as the hosts did.

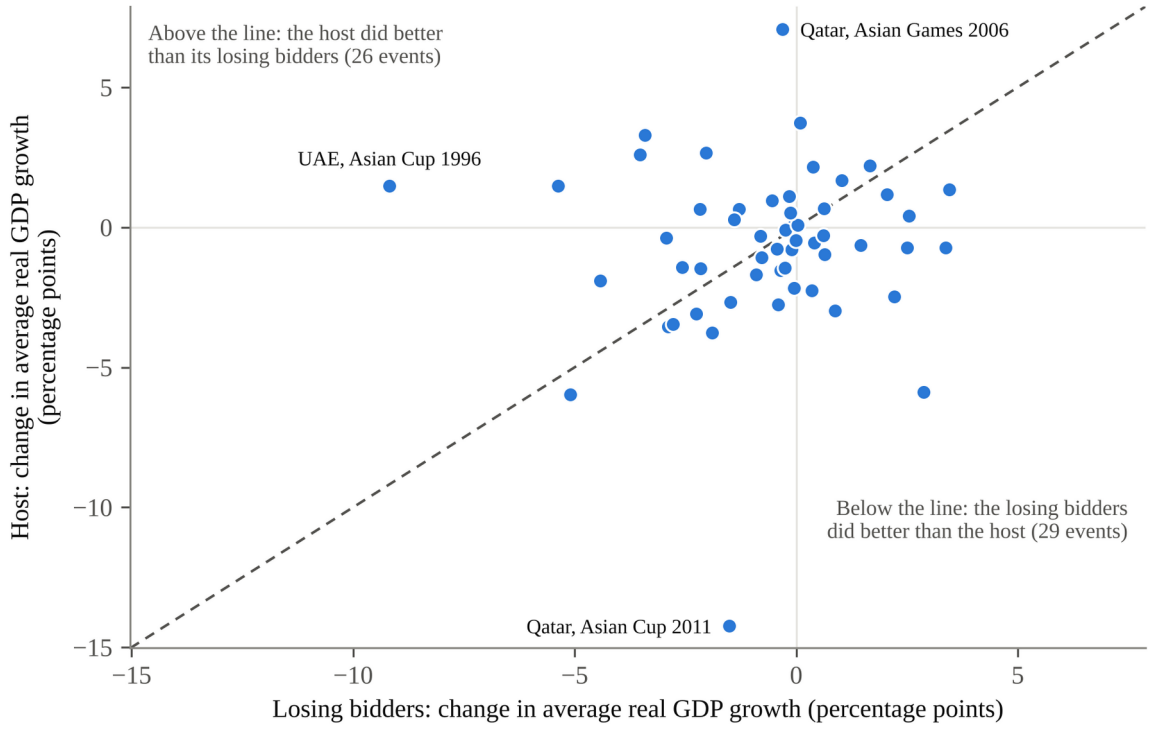


Figure 2. Change in average annual GDP growth, from the five years before each event to the five years after it, for the host and for the countries that bid for the same event and lost (55 events held from 1988 to 2019). The dashed line marks equal change; points above it are events in which the host did better than its losing bidders.

3.6 Jobs created per dollar spent

Many events report thousands or even hundreds of thousands of new jobs, but when those numbers are divided by cost, large events turn out to be an expensive way to create employment. The median number of jobs reported per $1 million spent was about 165 for Copa América, 88 for AFCON, and 76 for the AFC Asian Cup, compared with 35 for the Asian Games, 19 for the

Euros, 13 each for the World Cup and the Winter Olympics, and about 7 for the Summer Olympics. These figures come from organizers and include short-term and part-time positions, so they overstate lasting employment. Many of the positions are temporary jobs in construction or event operations, so the hiring that happens before an event does not necessarily lower unemployment in the long run.

3.7 Two contrasting cases: Qatar 2022 and Euro 2024

Qatar 2022 had the largest loss in my dataset, and Euro 2024 had one of the largest surpluses. Both cases also show why these numbers need to be read carefully. In my dataset, the 2022 World Cup in Qatar has an estimated cost of $220 billion and revenue of $7.6 billion, the largest gap of any event, and Qatar built seven new stadiums for it. The $220 billion figure includes metro lines, hotels, airports, and other projects in Qatar's national development plan rather than just the tournament, while the stadiums themselves were reported to cost $6.5 billion to $10 billion (Killingstad, 2022). The revenue figure is close to FIFA's revenue of about $7.5 billion for its whole 2019 to 2022 cycle, which is money that went to FIFA rather than to Qatar (Al Jazeera, 2022). Even with these corrections, Qatar spent far more than the tournament earned, but the case shows that a single headline number can exaggerate or hide what a host actually paid.

The 2024 UEFA European Championship in Germany is the opposite case. In my dataset it cost about $1 billion and earned $2.8 billion, and Germany built no new venues. UEFA reported record revenue of €2.5 billion from the tournament (Calcio e Finanza, 2025), but most of that revenue belongs to UEFA rather than to Germany, so the benefit to the host came mainly from visitor spending. A study that UEFA commissioned from Nielsen Sports estimated the tournament's total economic, media, and social impact on Germany and its ten host cities at €7.44 billion and put the cost of organizing the tournament, shared between UEFA's organizing company and the host cities, at about €650 million (Nielsen Sports, 2024). Because the organizer of the tournament paid for the study, its estimate is probably closer to the high end than the low end of what the event was worth. Taken together, the two cases show that whether an event pays off depends less on the sport than on how much the host has to build and on who collects the revenue.

3.8 What the averages miss: South Korea and Japan

When I discussed these results with my reviewer, we kept coming back to South Korea. The 1988 Seoul Olympics and the 2002 World Cup are often credited there, including by government agencies, with changing how the world saw the country and helping its economy grow, and the averages in this paper cannot fully test that claim. The 1988 Seoul Olympics are not in my dataset, whose Summer Olympics begin with Barcelona in 1992. South Korea's economy was also already growing very fast before the Seoul Games. According to IMF data, South Korea's real GDP grew by an average of 11.3 percent a year from 1983 to 1987 and 8.3 percent a year from 1989 to 1993 (International Monetary Fund, 2026). By the measure used in Section 3.5, South Korea would therefore look like one more host whose growth slowed: its growth fell by 3.0 percentage points, more than the 1.5-point drop in Japan, whose city of Nagoya lost the bid for the 1988 Games. What that measure cannot capture is whether the Games changed how other countries saw and dealt with South Korea. The National Archives of Korea (2015), a South Korean government agency, describes a country that the world had known mainly for the Korean War and its rapid development and that came to be seen in a new way because of the Seoul Olympics. The Soviet Union, which sent its team to Seoul in 1988 even though many people expected it to boycott the Games, established diplomatic relations with South Korea in 1990, and China followed in 1992 (Lankov, 2010). Because these ties came as the Cold War was ending, the Games were at most one of several causes. Rose and Spiegel (2009) point out that the decision to award the 1988 Games to South Korea coincided with the country's political liberalization. Their study measures trade rather than image or diplomacy, and it concludes that the rise in trade came from the signal that a country sends when it bids for the Games rather than from hosting them, which is why the countries that bid and lost gained as much as the hosts.

My own data give a mixed answer for South Korea's later events. The 2002 World Cup, which South Korea co-hosted with Japan, is one of the events in which the hosts did better than the country that lost the bid. South Korea's average growth fell from 5.4 percent a year in the five years before the tournament to 4.7 percent in the five years after it, while the growth of Mexico, the losing bidder, fell further, from 4.2 percent to 2.7 percent. That comparison says little about the World Cup itself, however. South Korea also hosted the 2002 Asian Games in Busan, and with the same South Korean growth figures, that event counts as one in which the

host did worse, because the growth of Taiwan, the losing bidder, rose. South Korea also did worse than its losing bidders after the 2014 Asian Games in Incheon and the 2018 Winter Olympics in PyeongChang. Both before averages for 2002 also include 1998, the worst year of the Asian financial crisis, when South Korea's economy shrank by 4.9 percent. Before the World Cup, the Korea Development Institute (2001), in a study first commissioned by the tournament's organizing committee, predicted that the boost to South Korea's international image would be very large, but it also noted that this kind of effect is difficult to measure with numbers. The tournament itself followed the pattern in Section 3.3: the two hosts built 17 new venues, and the tournament earned about 31 cents in revenue for every dollar it cost.

South Korea and Japan also show that what happens to a venue depends on who uses it afterward. South Korea built ten new stadiums for the 2002 World Cup. About a year after the tournament, the operators of the Seoul World Cup Stadium expected to turn 10.7 billion won in yearly revenue into a profit of 3.5 billion won, largely because the retailer Carrefour had agreed to pay 9.1 billion won a year in rent for a hypermarket inside the stadium. A multiplex cinema and a shopping mall had also opened there. The nine other new stadiums, by contrast, were not expected to make a profit that year (Lim, 2003). More than twenty years later, the Seoul stadium still depends on more than soccer. From January to August 2024, it earned 8.2 billion won from matches, concerts, and other events. Of that amount, 2.4 billion won came from concerts and other cultural events, including 1.4 billion won from concerts by the singer Lim Young-woong (Kim, 2024). Tokyo shows a similar contrast after the 2020 Games. Ariake Arena, which hosted volleyball during the Games, now holds concerts, and its operator has paid the Tokyo Metropolitan Government a larger profit-sharing payment each year since 2022. The Tokyo Metropolitan Government, by contrast, has paid the operating expenses of the Tokyo Aquatics Center every year (Tokyo Metropolitan Government, 2026). PyeongChang, which hosted the 2018 Winter Olympics, shows what happens when that kind of demand is missing. Gangwon Province spends about 6.3 billion won a year to keep six of its Olympic venues running and another 3 billion won a year on repairs. The four venues for sliding, ski jumping, cross-country skiing, and biathlon alone cost 5 billion won a year to operate, even though only about 30 national team athletes use them (Choi, 2025).

These cases explain why the averages come out the way they do. The international image that South Korean sources describe is something that a five-year GDP average cannot capture.

The Seoul World Cup Stadium stayed busy because a large city, a hypermarket, a cinema, and concert audiences gave it users after the tournament. Where that kind of demand did not exist, as in PyeongChang, the venues became the same kind of burden as the arenas in Rio de Janeiro. For a host government, the lesson is to plan each venue around the people who will use it after the event, not around the event itself.

4. Opportunity Cost

The money that hosts spend on these events could have been used elsewhere, and in some cases the trade-off was visible while the event was still being prepared. Across my dataset, the gap between reported costs and reported event revenue was about $493 billion, or about $281 billion without Qatar. Not all of that gap was public money, because it includes private investment and revenue that went to governing bodies such as FIFA, but much of it was: for London 2012 alone, the public sector funding package exceeded £9 billion (National Audit Office, 2007). Rio de Janeiro shows what that gap can mean for a single place. On June 17, 2016, less than two months before the Olympics, the acting governor of Rio de Janeiro state declared a state of financial emergency because the state was struggling to pay public salaries and pensions, and police departments and hospitals were complaining that they lacked basic supplies. The decree allowed the federal government to release about $850 million in emergency money (Play the Game, 2016). Six months after the Games, the state was still months late in paying teachers, hospital workers, and pensions, while the Olympic Park, which had cost about $800 million, sat largely empty (Wade, 2017). A government in that position appeared to face a direct trade-off between the Games and the services its residents rely on every day. The Rio case still cannot show what the state would have done with the money if it had not hosted, because the crisis also had other causes, including falling oil revenue, a national recession, and rising security spending (Play the Game, 2016). Comparing hosts with the countries that bid and lost is a stronger test, and Section 3.5 found no reliable difference between the two groups in GDP growth. The Rio case is therefore best read as an example of what the trade-off can look like for one government, not as proof of what hosting costs every country.

5. Limitations

These findings have several limits. First, GDP is affected by many factors that have nothing to do with sporting events, so the GDP results cannot prove what would have happened

without the event. The comparison with losing bidders makes that problem smaller, but the bidders are not a random group, since they also chose to bid, and the comparison covers only 55 of the 77 events, so it can detect only fairly large differences. The result does not depend on how the losing bidders are defined: keeping every country in my original candidate list gives 28 events in which the host did better and 32 in which it did worse ($p = 0.70$), and also leaving out bids that were withdrawn before the vote gives 25 and 29 ($p = 0.68$). Second, job estimates often include temporary positions, and they are usually reported by organizers, who have reasons to make them look large. Third, events report costs and profits using different accounting methods. Some cost figures include infrastructure that a country might have built anyway, and some revenue figures belong to governing bodies such as FIFA and UEFA rather than to the host. This is why I relied mainly on revenue-to-cost ratios and treated reported profits with caution. For the same reason, the dataset's totals depend heavily on how cost was defined for a few very large events, as the Qatar example shows, while the medians are much less sensitive to those choices. The totals also add dollars from different decades without adjusting for inflation, which is another reason to rely on ratios, since each ratio compares an event's revenue and cost from the same year. Because amounts are not adjusted for inflation, an older event may also fall into a lower cost group in Table 1 than it would in today's dollars. Costs and revenues for older and smaller tournaments are rough estimates as well, because those events published far less financial detail than recent Olympics and World Cups. Finally, the dataset does not measure the long-term value of new facilities, such as decades of community use, so a venue that looks like a loss in the year of the event could still produce benefits later, as the Seoul World Cup Stadium did.

6. Conclusion

Most of the events in my dataset did not pay for themselves. Events can create jobs, attract tourists, and bring in revenue, but for most hosts the costs are larger than the event's revenue, budgets rise well beyond the original projections, and new venues risk becoming expensive buildings with no tenant. Hosts' economies also tended to grow more slowly afterward, but so did the economies of the countries that bid and lost, and the data do not show that hosting made growth faster or slower. The clearest pattern in the data is that events held in facilities that already exist perform far better than events that require large amounts of new

construction. The experience of South Korea and Japan adds that when a host does build, the venues that stay busy are the ones with users waiting for them after the event, such as the Seoul World Cup Stadium with its hypermarket and concerts. It also shows that some possible benefits, such as a better international image, would not show up in growth figures even if they were real. A government deciding whether to bid should ask three questions: how much it will have to build, who will use those buildings afterward, and what the same money could have paid for instead.

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Appendix: The 91 Events in the Dataset

Costs and revenues are in millions of U.S. dollars of each event's own year. Amounts reported in euros or pounds were converted at about $1.10 per euro and $1.30 per pound, and the Beijing 2008 cost, reported as $40 billion to $44 billion, is shown at its midpoint. New venues counts newly built venues only, not renovated ones. Year follows the name of each edition: Tokyo 2020 and Euro 2020 were held in 2021, and AFCON 2023 was played in early 2024. Ratios were calculated before the costs and revenues were rounded for this table.

Event	Year	Host	Cost ($M)	Revenue ($M)	Revenue-to-cost ratio	New venues
Summer Olympics	1992	Barcelona, Spain	11,600	1,600	0.14	12
Summer Olympics	1996	Atlanta, USA	4,700	2,600	0.55	4
Summer Olympics	2000	Sydney, Australia	5,200	2,400	0.46	9
Summer Olympics	2004	Athens, Greece	15,000	2,200	0.15	14
Summer Olympics	2008	Beijing, China	42,000	3,300	0.08	12
Summer Olympics	2012	London, United Kingdom	16,800	3,800	0.23	8
Summer Olympics	2016	Rio de Janeiro, Brazil	23,600	2,700	0.11	10
Summer Olympics	2020	Tokyo, Japan	13,700	4,700	0.34	8
Summer Olympics	2024	Paris, France	7,100	4,710	0.66	2
Winter Olympics	1988	Calgary, Canada	1,100	510	0.46	5
Winter Olympics	1992	Albertville, France	1,200	600	0.50	7
Winter Olympics	1994	Lillehammer, Norway	1,100	530	0.48	10
Winter Olympics	1998	Nagano, Japan	2,200	1,100	0.50	6
Winter Olympics	2002	Salt Lake City, USA	2,500	1,900	0.76	3
Winter Olympics	2006	Torino, Italy	4,400	1,800	0.41	7
Winter Olympics	2010	Vancouver, Canada	7,200	2,600	0.36	5
Winter Olympics	2014	Sochi, Russia	51,000	3,100	0.06	11
Winter Olympics	2018	PyeongChang, South Korea	12,900	2,250	0.17	6
FIFA World Cup	1990	Italy	4,400	1,400	0.32	2
FIFA World Cup	1994	USA	500	550	1.10	0
FIFA World Cup	1998	France	2,300	360	0.16	1
FIFA World Cup	2002	Japan & South Korea	7,000	2,200	0.31	17
FIFA World Cup	2006	Germany	4,300	2,500	0.58	3
FIFA World Cup	2010	South Africa	3,600	3,560	0.99	5
FIFA World Cup	2014	Brazil	15,000	4,830	0.32	5
FIFA World Cup	2018	Russia	11,600	5,350	0.46	6
FIFA World Cup	2022	Qatar	220,000	7,570	0.03	7
UEFA Euro	1992	Sweden	38	52	1.35	0
UEFA Euro	1996	United Kingdom	165	168	1.02	0
UEFA Euro	2000	Belgium & Netherlands	495	265	0.54	1
UEFA Euro	2004	Portugal	4,180	215	0.05	7
UEFA Euro	2008	Austria & Switzerland	1,210	1,500	1.24	1
UEFA Euro	2012	Poland & Ukraine	34,100	1,600	0.05	6
UEFA Euro	2016	France	2,310	2,200	0.95	4
UEFA Euro	2020	11 cities in 10 countries	990	2,200	2.22	1
UEFA Euro	2024	Germany	1,045	2,800	2.68	0
Copa América	1989	Brazil	10	8.5	0.85	0
Copa América	1991	Chile	12	10	0.83	0
Copa América	1993	Ecuador	25	21	0.84	1
Copa América	1995	Uruguay	18	20	1.08	0
Copa América	1997	Bolivia	35	26	0.74	0
Copa América	1999	Paraguay	42	30	0.71	1
Copa América	2001	Colombia	50	42	0.84	0
Copa América	2004	Peru	65	54	0.83	0
Copa América	2007	Venezuela	1,200	145	0.12	5
Copa América	2011	Argentina	90	115	1.28	0
Copa América	2015	Chile	110	132	1.20	0
Copa América	2019	Brazil	140	185	1.32	0
Copa América	2021	Brazil	60	35	0.58	0
Copa América	2024	USA	207	319	1.54	0
Copa América Centenario	2016	USA	150	245	1.63	0

Event	Year	Host	Cost ($M)	Revenue ($M)	Revenue-to-cost ratio	New venues
AFCON	1990	Algeria	25	18	0.72	0
AFCON	1992	Senegal	40	28	0.70	1
AFCON	1994	Tunisia	45	38	0.84	0
AFCON	1996	South Africa	85	92	1.08	0
AFCON	1998	Burkina Faso	60	22	0.37	2
AFCON	2000	Ghana & Nigeria	90	65	0.72	0
AFCON	2002	Mali	140	35	0.25	5
AFCON	2004	Tunisia	95	82	0.86	1
AFCON	2006	Egypt	110	118	1.07	0
AFCON	2008	Ghana	160	125	0.78	2
AFCON	2010	Angola	1,000	140	0.14	4
AFCON	2012	Equatorial Guinea & Gabon	700	155	0.22	3
AFCON	2013	South Africa	60	162	2.70	0
AFCON	2015	Equatorial Guinea	60	110	1.83	0
AFCON	2017	Gabon	320	135	0.42	2
AFCON	2019	Egypt	140	187	1.34	0
AFCON	2022	Cameroon	850	142	0.17	3
AFCON	2023	Ivory Coast	1,000	98	0.10	4
AFC Asian Cup	1988	Qatar	15	10	0.70	0
AFC Asian Cup	1992	Japan	65	52	0.80	1
AFC Asian Cup	1996	United Arab Emirates	95	81	0.85	3
AFC Asian Cup	2000	Lebanon	110	45	0.41	2
AFC Asian Cup	2004	China	185	142	0.77	1
AFC Asian Cup	2007	Indonesia, Malaysia, Thailand & Vietnam	160	115	0.72	0
AFC Asian Cup	2011	Qatar	210	175	0.83	0
AFC Asian Cup	2015	Australia	120	143	1.19	0
AFC Asian Cup	2019	United Arab Emirates	340	295	0.87	1
AFC Asian Cup	2024	Qatar	45	312	6.96	0
Asian Games	1990	Beijing, China	530	410	0.77	16
Asian Games	1994	Hiroshima, Japan	1,300	1,100	0.85	8
Asian Games	1998	Bangkok, Thailand	650	510	0.78	5
Asian Games	2002	Busan, South Korea	2,100	1,800	0.86	12
Asian Games	2006	Doha, Qatar	2,800	620	0.22	9
Asian Games	2010	Guangzhou, China	18,000	1,500	0.08	12
Asian Games	2014	Incheon, South Korea	1,620	860	0.53	16
Asian Games	2018	Jakarta and Palembang, Indonesia	2,400	1,100	0.46	4
Asian Games	2023	Hangzhou, China	14,500	1,300	0.09	12
UEFA Champions League Final	2025	Germany	14	58	4.19	0
The Open (golf)	2025	United Kingdom	134	200	1.49	0
Wimbledon	2025	United Kingdom	458	542	1.18	0